

Lab Evaluation:

Course Name: Object Oriented Programming				Course Code: CMPE221L		
Assignment Type: Lab				Dated: 17 th October 2025		
Semester: 3 rd				Session: CYS 2025 Spring		
Lab/Project/Assignment #: 3				CLOs to be covered: CLO 1		
Lab Title: Getting Started with JAVA				Teacher Name: Engr. Afeef Obaid		
CLO 1	Implement abstraction and encapsulation to develop reusable classes for objects of real-world problems					
Levels (Marks)	Level 1	Level 2	Level 3	Level 4	Level 5	Level 6
(10)						
Total						/10

Rubrics for Current Lab Evaluation

Scale	Marks	Level	Rubric
Excellent	9-10	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	7-8	L2	Submitted the lab tasks but have good understanding
Good	5-6	L3	Submitted the lab tasks but have weak understanding.
Basic	3-4	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1-2	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No. 3

Lab Goals/Objectives:

- Getting Started with Java
- Acquiring practice with familiar programming constructs in JAVA

Equipment Required: Intel J IDEA Edition Community, Net Beans with JDK 13 Version, Visual Studio.

Theory/Relevant Material: Write First Java Program

1. Modify the **Point** class as discussed in class: that can represent 3-D point. Modify all the existing functions such that they work for 3-D point as well.
2. Think of **Time** as an entity. Think of attributes and functions. Implement getter setters for its attributes. And write a function *whatTime(int no_of_seconds_till_noon)* and sets its attributes accordingly.
3. You have graduated from University and now have a great job! You want to save for buying some physical asset (like house, car, Alienware or macbook etc). Now assets are never cheap, you realize you are going to have to save for several years before you can afford to make the down payment on it. We want to determine how long it will take you to save enough money to make the down payment given the following assumptions:
 - i. Call the cost of your dream asset **total_cost**.
 - ii. Call the portion of the cost needed for a down payment **portion_down_payment**. For simplicity, assume that $\text{portion_down_payment} = 0.25$ (25%) of the **total_cost**.
 - iii. Call the amount that you have saved thus far **current_savings**. You start with a current savings of \$0.
 - iv. Assume that you invest your current savings wisely, with an annual return of **r** (in other words, at the end of each month, you receive an additional $\text{current_savings} * r / 12$ funds to put into your savings – the 12 is because **r** is an annual rate). Assume that your investments earn a return of $r = 0.04$ (4%).
 - v. Assume your monthly salary is **monthly_salary**.
 - vi. Assume you are going to dedicate a certain amount of your salary each month to saving for the down payment. Call that **portion_saved**. This variable should be in decimal form (i.e. 0.1 for 10%).
 - vii. At the end of each month, your savings will be increased by the return on your investment, plus a percentage of your monthly **salary** (annual salary / 12).

Write a program to calculate how many months it will take you to save up enough money for a down payment. You will want your main variables to be floats, so you should cast user inputs to floats.

Your program should ask the user to enter the following variables:

1. The starting monthly salary (**monthly_salary**)
2. The portion of salary to be saved (**portion_saved**)
3. The cost of your dream home (**total_cost**)

Try different inputs and see how long it takes to save for a down payment.

Sample input

>>>

Enter your monthly salary: 10000

Enter the percent of your salary to save, as a decimal: .10 Enter the cost of your dream home: 1000000

Number of months: 183

>>>

Enter your monthly salary: 6666.66

Enter the percent of your salary to save, as a decimal: .15 Enter the cost of your dream home: 500000

Number of months: 105

>>>